Lesson 7.2 - Cumulative Distribution Functions and Probability Density Functions

> **Video:** Cumulative Distribution Functions and Probability Density Functions by The Organic Chemistry Tutor **Chapter:** 7 - Continuous Distributions

PDF vs CDF

	PDF, probability density function	**CDF**, cumulative distribution function
Notation	$f(x)$	$F(x) = P(X \leq x)$
Tells you	The **shape** of the distribution and the curve's **height** at x	The **area to the left** of x, the probability accumulated up to x
Range	$f(x) \geq 0$	$0 \leq F(x) \leq 1$

The total area under any PDF is 1.

Uniform distribution on $[a, b]$

PDF (a constant height; no x in it):

$$f(x) = \frac{1}{b-a}$$

CDF (the area of the rectangle from a to x = base × height):

$$F(x) = P(X \leq x) = (x-a) \cdot \frac{1}{b-a} = \frac{x-a}{b-a}$$

Exponential distribution

PDF (it starts at height λ and decays, with $\lambda = 1/\mu$):

$$f(x) = \lambda e^{-\lambda x}$$

CDF (area to the left):

$$F(x) = P(X \leq x) = 1 - e^{-\lambda x}$$

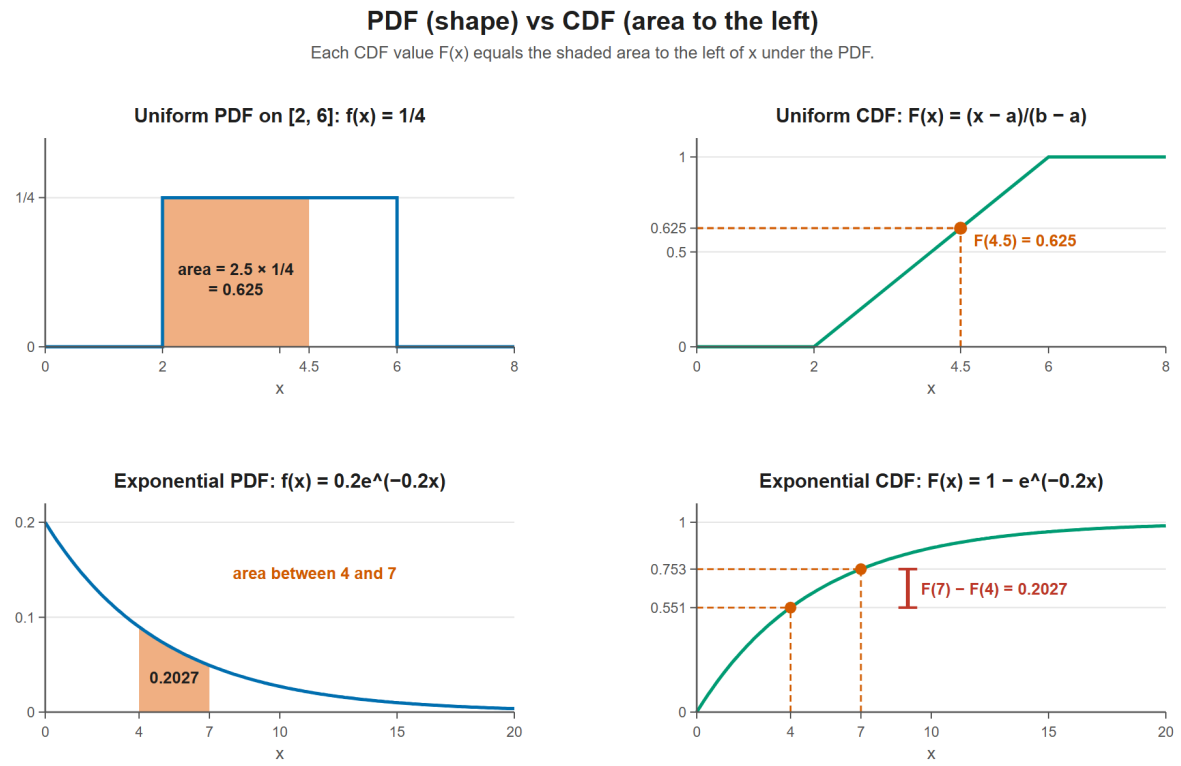
Area to the right (not a CDF, because the CDF is always the left area):

$$P(X \geq x) = 1 - F(x) = 1 - (1 - e^{-\lambda x}) = e^{-\lambda x}$$

Area between a and b: subtract two left areas.

$$P(a \leq X \leq b) = F(b) - F(a) = (1 - e^{-\lambda b}) - (1 - e^{-\lambda a}) = e^{-\lambda a} - e^{-\lambda b}$$

Picture: the region left of b minus the region left of a leaves the region between a and b.



Left column: the PDF, whose shaded area is a probability. Right column: the CDF, whose height at x is the area to the left of x. The area between 4 and 7 under the exponential PDF (0.2027) equals the jump F(7) − F(4) on its CDF.

Reminders for continuous distributions

- $P(a \leq X \leq b) = P(a < X < b)$, because the endpoints don't change the area.
- $P(X = a) = 0$: a single line has height but no width.

Key takeaways

- **PDF** $f(x)$ gives the **shape and height**. **CDF** $F(x)$ gives the **area to the left** (cumulative probability).
- Right area = $1 - F(x)$. Area between = $F(b) - F(a)$.
- Uniform: $F(x) = \dfrac{x - a}{b - a}$. Exponential: $F(x) = 1 - e^{-\lambda x}$.